

Master Thesis: Proposal

Student name: *Dominik Künkele*

Course: *Master project (LT2402)*

Due date: *January 30th, 2023*

Tentative title: On the table or on the shelf: Teaching a situated agent spatial relations

Student's name: Dominik Künkele (dominik.kuenkele@outlook.com, +49 151 56043266)

Supervisor: Simon Dobnik

Examiner: Asad Sayeed

Description and Motivation

While current models are getting better at grasping entities and actions of visual scenes, they still struggle to understand spatial relations between those entities. Models don't need to only understand their geometrical, but also their functional and pragmatic relations. An umbrella can for example be *over* a person if it's protecting them from rain even though it's geometrically not directly above them (functional) [1].

Furthermore, referring to spatial relations depend on perspectives the speaker take [2], [3]. Speakers can for example take an egocentric view to describe the scene from their perspective, or they can take an allocentric view to describe it from another perspective. In both cases, a model would need to understand, which is used to resolve the ambiguity of the reference.

When the model is used by a situated agent like a robot, the understanding of spatial relations gets even more important, since a core part is to interact directly with their environment. For this, they need this kind of knowledge to navigate or interact with objects and persons.

The goal of the Master Thesis is to explore the understanding of spatial relations and their ambiguity deeper. It is not yet fixed, how exactly this will be done. One topic may be to interactively teach a robot to 'put a mug on the table' and 'put a mug on the shelf'. The final location in both cases would be different depending on the object involved. Therefore, the robot would need to understand the *function* of the object (i.e. table, shelf) to derive the meaning of the *spatial relation* (i.e. on the). This experiment would be done in *Habitat Lab* [4], a 3D-world simulation, where the developer has control over the complete environment and a simulated robot can perceive this environment with different sensors. The work would include the creation of an artificial dataset in the 3D-world as well as teaching the robot the spatial relations based on this dataset.

Required knowledge and skills

- Python
- Image processing
- Machine Learning
- 3D simulations
- Dialogue handling

List of resources

- Habitat Lab [4]/Unity Engine
- existing 3D worlds and objects (e.g. Habitat AI templates) [4]
- Python ML libraries

Work plan

Start date	Work packages	duration
30 th January	• reading background in papers • getting familiar in 3D environments (Habitat Lab, Unity)	2 weeks
13 th February	• reading about teaching situated agents in dialogue • defining teaching/dialogue framework	2 weeks
27 th February	• creating 3D scenes dataset • defining scope of experiments • writing theoretical background and motivation	3 weeks
20 th March	• teaching experiments • analyzing results • adapting parameters	4 weeks
17 th April	• writing	3 weeks
8 th May	• submission to supervisor • revising and finishing	2 week

17 th May	• submission for defense	
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References

- [1] Kenny R. Coventry, Angelo Cangelosi, Rohanna Rajapakse, Alison Bacon, Stephen Newstead, Dan Joyce, and Lynn V. Richards. Spatial prepositions and vague quantifiers: Implementing the functional geometric framework. In *Spatial Cognition IV. Reasoning, Action, Interaction*, pages 98–110. Springer Berlin Heidelberg, 2005.
- [2] Jae Hee Lee, Matthias Kerzel, Kyra Ahrens, Cornelius Weber, and Stefan Wermter. What is right for me is not yet right for you: A dataset for grounding relative directions via multi-task learning. May 2022.
- [3] L. Steels and M. Loetzsch. Perspective alignment in spatial language. May 2006.
- [4] Andrew Szot, Alex Clegg, Eric Undersander, Erik Wijmans, Yili Zhao, John Turner, Noah Maestre, Mustafa Mukadam, Devendra Chaplot, Oleksandr Maksymets, Aaron Gokaslan, Vladimir Vondrus, Sameer Dharur, Franziska Meier, Wojciech Galuba, Angel Chang, Zsolt Kira, Vladlen Koltun, Jitendra Malik, Manolis Savva, and Dhruv Batra. Habitat 2.0: Training home assistants to rearrange their habitat. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.